

Indexing Report Project: ULK Academic Activity and Resource Management System Phase 11: Indexing Experiment

1. 1. Objective The goal of this experiment is to measure the performance impact of creating a B-tree index on a frequently searched column.

2. 2. Test Environment

- **Database:** PostgreSQL 16
- **Table:** Student
- **Total Records:** 100
- **Test Query:** `SELECT * FROM Student WHERE student_code = "STU0050";`

3. 3. Performance Results

- **Without Index:** Sequential Scan (0.250 ms)
- **With Index:** Index Scan (0.045 ms)
- **Improvement:** ~82% faster